

1. Data, Measurements & Representations

1.1 Paradigms: Frequentist, Bayesian, Stochastic

1.2 Recap Calculus

2. Anomaly Detection

2.1 PCA

2.2 GMMs & EM algo

2.3 Summary

3. Density Estimation

3.1 Parametric Inference

3.2 Bayesian Inference

3.3 Linear Models

3.4 Kernel Regression

3.5 Linear Models

3.6 Linear Models

3.7 Linear Models

3.8 Linear Models

3.9 Linear Models

3.10 Linear Models

3.11 Linear Models

3.12 Linear Models

3.13 Linear Models

3.14 Linear Models

3.15 Linear Models

3.16 Linear Models

3.17 Linear Models

3.18 Linear Models

3.19 Linear Models

3.20 Linear Models

3.21 Linear Models

3.22 Linear Models

3.23 Linear Models

3.24 Linear Models

3.25 Linear Models

3.26 Linear Models

3.27 Linear Models

3.28 Linear Models

3.29 Linear Models

3.30 Linear Models

3.31 Linear Models

3.32 Linear Models

3.33 Linear Models

3.34 Linear Models

3.35 Linear Models

3.36 Linear Models

3.37 Linear Models

3.38 Linear Models

3.39 Linear Models

3.40 Linear Models

3.41 Linear Models

3.42 Linear Models

3.43 Linear Models

3.44 Linear Models

3.45 Linear Models

3.46 Linear Models

3.47 Linear Models

3.48 Linear Models

3.49 Linear Models

3.50 Linear Models

3.51 Linear Models

3.52 Linear Models

3.53 Linear Models

3.54 Linear Models

3.55 Linear Models

3.56 Linear Models

3.57 Linear Models

3.58 Linear Models

3.59 Linear Models

3.60 Linear Models

3.61 Linear Models

3.62 Linear Models

3.63 Linear Models

3.64 Linear Models

3.65 Linear Models

3.66 Linear Models

3.67 Linear Models

3.68 Linear Models

3.69 Linear Models

3.70 Linear Models

3.71 Linear Models

3.72 Linear Models

3.73 Linear Models

3.74 Linear Models

3.75 Linear Models

3.76 Linear Models

3.77 Linear Models

3.78 Linear Models

3.79 Linear Models

3.80 Linear Models

3.81 Linear Models

3.82 Linear Models

3.83 Linear Models

3.84 Linear Models

3.85 Linear Models

3.86 Linear Models

3.87 Linear Models

3.88 Linear Models

3.89 Linear Models

3.90 Linear Models

3.91 Linear Models

3.92 Linear Models

3.93 Linear Models

3.94 Linear Models

3.95 Linear Models

3.96 Linear Models

3.97 Linear Models

3.98 Linear Models

3.99 Linear Models

3.100 Linear Models

6. Model Validation & Uncert. Quant

6.1 Statistical Learning & Model Val.

6.2 Cookbook

6.3 Uncertainty Quantification with Linear Methods

6.4 Bayesian Neural Networks

6.5 Active Learning

6.6 Batch Active Learning

6.7 Convex Optimization

6.8 SVMs

6.9 SVMs

6.10 SVMs

6.11 SVMs

6.12 SVMs

6.13 SVMs

6.14 SVMs

6.15 SVMs

6.16 SVMs

6.17 SVMs

6.18 SVMs

6.19 SVMs

6.20 SVMs

6.21 SVMs

6.22 SVMs

6.23 SVMs

6.24 SVMs

6.25 SVMs

6.26 SVMs

6.27 SVMs

6.28 SVMs

6.29 SVMs

6.30 SVMs

6.31 SVMs

6.32 SVMs

6.33 SVMs

6.34 SVMs

6.35 SVMs

6.36 SVMs

6.37 SVMs

6.38 SVMs

6.39 SVMs

6.40 SVMs

6.41 SVMs

6.42 SVMs

6.43 SVMs

6.44 SVMs

6.45 SVMs

6.46 SVMs

6.47 SVMs

6.48 SVMs

6.49 SVMs

6.50 SVMs

6.51 SVMs

6.52 SVMs

6.53 SVMs

6.54 SVMs

6.55 SVMs

6.56 SVMs

6.57 SVMs

6.58 SVMs

6.59 SVMs

6.60 SVMs

6.61 SVMs

6.62 SVMs

6.63 SVMs

6.64 SVMs

6.65 SVMs

6.66 SVMs

6.67 SVMs

6.68 SVMs

6.69 SVMs

6.70 SVMs

6.71 SVMs

6.72 SVMs

6.73 SVMs

6.74 SVMs

6.75 SVMs

6.76 SVMs

6.77 SVMs

6.78 SVMs

6.79 SVMs

6.80 SVMs

6.81 SVMs

6.82 SVMs

6.83 SVMs

6.84 SVMs

6.85 SVMs

6.86 SVMs

6.87 SVMs

6.88 SVMs

6.89 SVMs

6.90 SVMs

6.91 SVMs

6.92 SVMs

6.93 SVMs

6.94 SVMs

6.95 SVMs

6.96 SVMs

6.97 SVMs

6.98 SVMs

6.99 SVMs

6.100 SVMs

time  $|C| < \infty$  and  $R(C) = 0$ . Then  
 $\eta \in E$  and it holds:  
 $\eta(C) \leq |C| \leq |C|$   
 $\eta(C) \leq \frac{1}{2} \log(|C|)$   
 $P(R(C) > 0) \leq P(\max_{C \in \mathcal{R}(\eta)} R(C) > 0)$   
 $\frac{1}{2} \log(R(C)) \leq \frac{1}{2} \log(|C|)$   
 $P(R(C) = 0) \leq |C| \cdot (1 - e^{-|C|}) \leq |C| e^{-|C|}$   
 $\int_0^\infty P(X > t) dt = \int_0^\infty 1 - F(t) dt = \int_0^\infty 1 - F(t) dt$   
 $\int_0^\infty P(X > t) dt = \int_0^\infty 1 - F(t) dt = \int_0^\infty 1 - F(t) dt$

$$R(\hat{c}_n) = \hat{R}_n(c^*) - R(c^*)$$

$R(c) - R(c) \sup_{c \in \mathcal{C}} R(c) - R(c)$   
 inform conv. "perfect" on positive conv.  
 $\inf_{c \in \mathcal{C}} R(c) - R(c)$   
 infimize  $R(c) \in \mathcal{C} \sup_{c \in \mathcal{C}} R(c) - R(c) \geq \frac{\epsilon}{2}$   
 nighst 2 cases  
 $c$  selected before seeing  $D_{\text{train}}$ . Then  
 cl. classification risk of  $c$  conv.  
 $c$  expected classification risk (ULN)  
 $E_{\mathcal{D}}$  selected according to an induction  
 (ERM) after we have seen the data  $\mathcal{X}$   
 not apply,  $R_{\mathcal{D}}(c) \leq 2(\epsilon/2)$   
 $X \geq 0 \Rightarrow \mathbb{P}(X \geq t) \leq \frac{E(X)}{t}$

$$\mathbb{E}[e^{a(X-b)}] \leq e^{a^2(b-a)^2/8}$$

Thm: Let  $X_1, \dots, X_n$  be indep bounded RVs  
 $|X_i| \leq b_i$  &  $S_n = \sum_{i=1}^n X_i$ . Then  $\forall t > 0$  :

$$P(S_n \geq t) \leq \exp\left(-\frac{t^2}{\sum_{i=1}^n (b_i - a_i)^2}\right)$$

1) Markov with  $P(\exp(X)) \leq \exp(c)$ , 2) i.i.d.

$$S = \sum_{i=1}^n (b_i - a_i)^2$$
$$\Rightarrow P(|\bar{Z}_n - E_x[\bar{Z}_n]| \geq \varepsilon) \leq 2 \exp\left(-\frac{2n\varepsilon^2}{\sum_{i=1}^n (b_i - a_i)^2 P_i/n}\right)$$

$$Y \Rightarrow P\left(\sup_{c \in \mathcal{C}} |\hat{R}_n(c) - R(c)| > \epsilon\right) \leq 2N_{\text{exp}}(2n\epsilon^2)$$

$$N_{\text{exp}}(2n\epsilon^2) \Rightarrow \epsilon = \sqrt{\frac{\ln(2N) - \ln(5\delta)}{2n}}$$
 "variance"

$x_1, x_0$  set of hyperplanes. Select  $(\hat{n})$   
 & extn error:  $\#C = 2 \cdot (n)$   
 in  $\hat{R}_n(c) \Rightarrow \forall c \in C: \hat{R}_n(c) \geq \hat{R}_n(c^*) - \frac{d}{n}$   
 $\hat{R}_n(c^*)$   
 $X$  has a density. If  $c^*$  found by emp.

ation, then,  $\forall$  dist of  $(X, Y)$ , if  $n \geq 1$  &  $\frac{d}{2n} \leq 1$   
 $\forall \epsilon \in (0, 1) \exists \delta \in (0, \epsilon) \exists c \in (0, 1) \exists n_0 \in \mathbb{N}$  s.t.  $n \geq n_0$  &  $\frac{d}{2n} \leq \delta$  implies  $P_n(\epsilon) \geq \inf_{(X, Y)} P_n(\delta) + c \in (0, 1)$

**Remarks**

$$= \max\{0, 1 - y, z\}, z \in \{-1, 1\}$$

Criteria:  $M_{PSO} \rightarrow$  all upper left  
minors have positive det.  $(k \leq n)$

let  $\mathcal{C}$  be a finite concept class & assume

Let  $\mathcal{A}$  be an algorithm that returns Hypothesis  $(\text{Value}: \mathcal{R}(Z) = 0)$  for any target  $c \in \mathcal{A}$  and iid sample  $Z$ . For any  $\epsilon, \delta > 0$ ,  $\log(\frac{1}{\epsilon}) \cdot \log(\frac{1}{\delta})$ , then success prob. is  $\geq 1 - \delta$  or  $\mathbb{P}(\mathcal{R}(Z) \geq \epsilon) \geq 1 - \delta$

Input processing:    $\leq 8$

Diagram illustrating a Variational Autoencoder (VAE) architecture. The input  $x$  is processed by an encoder (Auto + Noise) to produce a latent variable  $z$ . This  $z$  is then processed by a decoder (Auto + Noise) to produce the output  $\hat{x}$ . The diagram also includes handwritten notes: "self-att" and "Masked A.H." for the encoder, "VAE decoder" for the decoder, and "Forward process" with the differential equation  $dx = S + du$  and the probability formula  $p(x | (z, u)) = \prod_i p(x_i | (z, u_i))$  for the forward process. The output is labeled "Output probability".

[illegible]